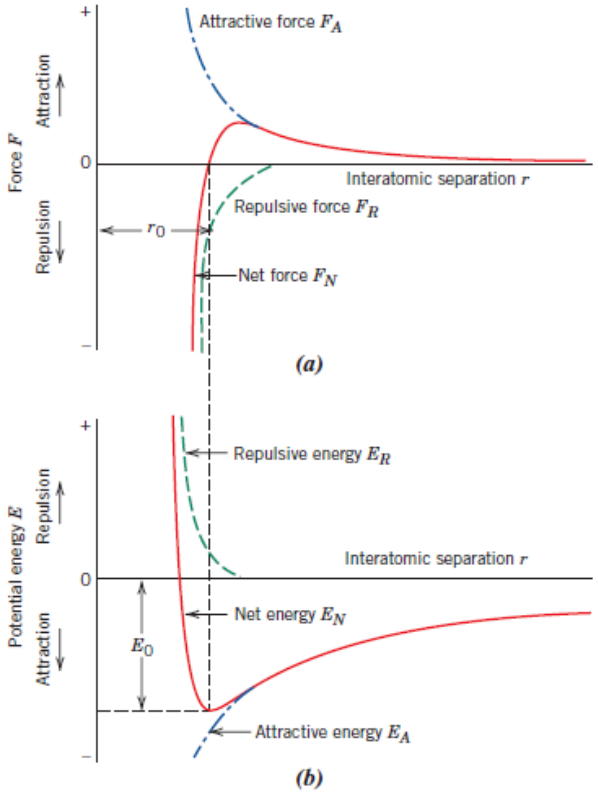
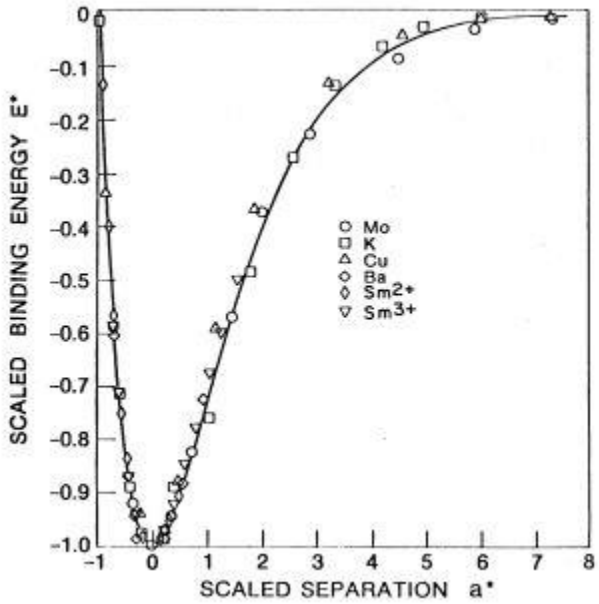
# EE2021: Tutorial -1

1. Show mathematically that the existence of molecules, compounds and solids was possible only because the attractive force between atoms (neutral or charged) varies more slowly with interatomic distance than the repulsive force between them




2. Imagine that you are a scholar doing research on a new metal-*Unobtanium*. Your advisor had told you that potential energy vs distance for most metals lie on the same curve if properly scaled/normalized as done above (right). Please come up with a scheme to normalize your experimental data on Unobtanium and verify whether your experimental data is indeed true.